

A Framework for Experimentation Pipelines with LLM Simulators

June 18, 2026*

Summary

This note accompanies the application “Scientific Discovery, Experimentation and AI Simulations” for the Schmidt Sciences AI at Work program. The note provides additional details on the theoretical framework for quantifying the value of information of AI simulators given a set of historical experiments for decision makers deciding which treatments to test and ultimately deploy. The goal is to formalize how to think about this problem in the context of Empirical Bayes methods, how to utilize historical experiments to calibrate and evaluate the simulations and, ultimately, how to design what information researchers should be given in a “simulation brief” to improve the decision making process.

The note is organized as follows. The rest of this section describes the motivation for our proposal and summarizes our project. Section 1 describes the theoretical set up for the experimentation pipeline and provides a real example from Airbnb’s ML experimentation team. Section 2 formalizes the decision maker’s objective, the Empirical Bayes framework and present our measure of value of information. Section 3 describes the AI simulator design, the proposed simulation brief that is our treatment and provides examples. Appendix sections A-G provide additional theoretical details for the screening problem and simulations to investigate which AI simulators should be built depending on the outcomes and welfare functions.

Introduction. Potentially the most important question about the impact of AI is whether it will improve the rate of scientific discovery. At a high level, this is what this proposal is about. The scientific discovery process is not just about running experiments. It is also about deciding which hypotheses need to be tested and which ones do not. Across domains, scientists and research teams face the same problem: ideas are abundant, but experiments capacity is not. Clinical trials, lab experiments, development RCTs, product A/B tests, all require teams to choose a small number of hypotheses from a much larger set of plausible ones. Testing the wrong hypotheses, or in the wrong order, wastes scarce experimental capacity, makes finding the best possible policy harder and ultimately delays scientific discovery. These prioritization decisions are often made using expert judgment and theoretical devices. Experts form priors about which ideas are promising, and which mechanisms are plausible. But these priors and theoretical devices are also costly to produce and may not be very informative in some cases (e.g. disagreement over the right structural model). Generative AI offers a new possibility to help streamline this process: before running a costly experiment, teams may be able to use LLM-based simulations to generate cheap, structured, pre-experimental evidence about how a proposed treatment might perform.

*This note was partially drafted with the help of Claude Code and ChatGPT Pro. All errors are our own.

The challenge is that raw LLM simulations cannot be treated as ground truth. LLM simulations may be biased, insensitive to heterogeneity, or unreliable outside familiar domains. So the question is: can gen AI simulations help scientists allocate experimental capacity better?

Our proposal is twofold. First, we develop an empirical Bayes theoretical framework and open source tool to help researchers, policy makers, and scientific teams understand the information value of gen AI simulations in experimentation settings with cost constraints. This note serves as grounding for the theoretical framework and describes how to build the simulator. Given a set of historical experiments, a new treatment we want to test, a human prior of its effect, and a utility function over outcomes including costs, we construct the posterior effect distribution given the AI simulation. This helps understand when the AI simulation forecasts are informative, when they are biased, when they are out of support, and, importantly, the expected information value of running a real or pilot experiment.

Second, we embed this tool in a real, high throughput, high stakes, experimentation funnel at a tech company (Airbnb). An environment in which ideas are well documented, the scientific discovery funnel is well defined, from idea generation to small pilots, to large AB tests to eventual deployment, the hypothesis space is large and the cost constraints are meaningful. This note also gives examples of the experimentation tunnel and how the simulation briefs might look like. Different teams will receive in a staggered fashion access to our simulation tool that given their candidate hypothesis documents will provide experimental briefs describing the expected treatment effects, information value of running a small pilot and calibration quality across different outcomes. We will then measure how teams change their approach to what to pilot and deploy. We are primarily interested in understanding first if AI simulations speed up discovery, by making the funnel shorter (time to deployment), whether they help prioritize the right pilots (that is the ones that offer the most information gain), and ultimately whether the deployed ideas are more successful meaning we are discovering better treatments. Finally, we will also survey the teams to understand qualitatively how their thought processes were shaped by access to the simulations. Below we provide more detail on the proposed pipeline and target output.

Proposal. The proposal is to combine an LLM simulator with a decision-theoretic selection rule. For each candidate, the simulator predicts the effect on the primary and guardrail outcomes the firm cares about. Those raw predictions are calibrated against past experiments, producing an estimate of the candidate’s expected welfare gain together with the uncertainty around that estimate. The selection rule then compares two welfare gains for every candidate. The first is the expected welfare gain from acting now on the calibrated belief alone, by deploying the candidate or by dropping it. The second is the expected welfare gain from first running a pilot of size n , observing its result, and only then choosing whether to deploy, net of pilot cost. Whichever expected gain is higher determines the action. Candidates whose calibrated welfare clears the deployment threshold are launched without a pilot, candidates whose calibrated welfare sits clearly below are dropped, and candidates near the threshold are routed to the pilot, at the size that maximizes net welfare gain.

Output. The procedure produces three outputs: an AI simulator and calibration recipe, a per-candidate simulation brief, and a measurable improvement in how pilots are allocated and policies deployed.

The simulator is run on each candidate using only information that was available at the time of evaluation, so its predictions are not contaminated by knowledge of future outcomes. A historical panel $\mathcal{D}_H$ stores past raw simulator predictions alongside their realized RCT estimates and proposal features. An empirical-Bayes calibration model fitted on $\mathcal{D}_H$ then turns the raw simulator output

for any new candidate into a calibrated posterior on its treatment effects, with pilot sampling noise correctly removed from the calibration residual so that calibration uncertainty is not confounded with pilot noise.

For each candidate, the simulation brief reports the calibrated mean and uncertainty on the welfare gain, the probability of clearing the deployment threshold, calibration diagnostics drawn from the most similar past experiments, the expected variance reduction at each feasible pilot size, and the recommended action of launching, dropping, or piloting the candidate at the corresponding optimal pilot size.

The expectation is that if the tool is useful in day-to-day decision making pilot capacity stops being spent on candidates whose calibrated belief already supports a clear launch or a clear stop, and instead is allocated to the marginal candidates where a pilot can change the decision. Every recommendation is tied to a calibrated posterior, so the gap between simulator predictions and realized pilot estimates is recorded and the calibration map is retrained as $\mathcal{D}_H$ grows, allowing us to inspect the change in behavior and retroactively assess the value of historical experiments. The simulator’s contribution to decision quality is then itself a measurable quantity, together with how it shapes what hypothesis are tested, the speed of testing and, ultimately, the success of the deployed policies.

1 Set up

We are interested in modeling the experimentation pipeline, from idea generation to policy deployment. The standard pipeline is comprised of different stages, each of which is characterized by a different set of decisions and information. At a high level, a standard experimentation funnel is comprised of the following stages:

1. Idea generation: The decision maker generates a set of candidate policies, or treatments, to choose from.
2. Pilot experiments: The decision maker runs a set of pilot experiments to evaluate a subset of candidate policies.
3. Policy learning: The decision maker uses the pilot experiments to learn the effect of the candidate policies on the outcomes and then chooses a subset of policies to test in a larger scale experiment.
4. Policy deployment: The decision maker deploys the best policy given the results of the experiments.

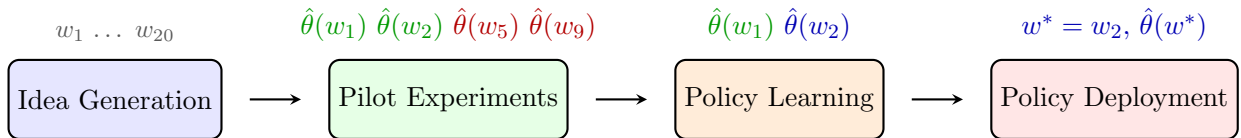


Figure 1: Experimentation funnel: candidate policies w_1 – w_{20} (gray), filtered to w_1, w_2, w_5, w_9 in pilots, then w_2 (blue) selected for deployment after a large RCT.

At the policy deployment time, the decision maker evaluates treatments on a vector of outcomes. Using the potential outcomes framework, the effect of policy $w \in \mathcal{W}$ on outcome k is defined by

$$\theta(w) = \begin{pmatrix} \theta_1(w) \\ \vdots \\ \theta_K(w) \end{pmatrix},$$

where the components may include treatment effects on outcomes relative to a baseline policy. For example,

$$\theta_k(w) = \mathbb{E}[Y_k(w) - Y_k(0)],$$

where $Y_k(w)$ is the potential outcome of outcome k under policy w and $Y_k(0)$ is the potential outcome of outcome k under the baseline policy (i.e. the status quo). Y_k may denote a primary outcome, such as bookings, revenue, retention, or conversion, or a guardrail outcome, such as safety, trust, cancellations, support contacts, fairness, or marketplace balance.

The decision maker has a utility function $U : \mathbb{R}^K \rightarrow \mathbb{R}$ that maps the treatment effect vector to a scalar utility. The decision maker’s objective is to choose the treatment that maximizes their utility.

$$w^* \in \arg \max_{w \in \mathcal{W}} U(\theta(w)).$$

Two classes of problems make this objective hard for the decision maker. First, the decision maker does not directly observe the true $\theta(w)$, so it needs to be estimated. Second, the decision maker may be subject to implementation constraints that limit the set of policies that can be implemented.

To address the first problem, traditionally, the decision maker would generate evidence about the true $\theta(w)$ by running experiments. RCT estimates provided unbiased estimates of different $\theta(w)$ for different policies, which can be used to choose the optimal policy. However, RCTs are expensive and time-consuming to run, and may not be feasible for all policies. Figure 1 shows an example of the funnel. The researcher has 20 candidate treatments at the start, but can only pilot a few. After the pilot stage, the researcher discards the treatments that are not promising and compares the two final candidates in a large RCT. Finally, on the basis of the RCT results, the researcher chooses the optimal policy w_2 .

However, this approach is not scalable. The decision maker may not have the resources to run a large RCT for each candidate treatment. Additionally, the decision maker may not have the time to run a large RCT for each candidate treatment. What would have happened if the decision maker could have tested all 20 treatments in the pilot stage? Figure 2 shows a framework for experimentation with a LLM simulator, in which the decision maker can simulate all 20 treatments in simultaneously in the pilot stage. The LLM simulator provides a prediction $\tilde{\theta}(w)$ for each candidate treatment, which can be used in combination with the pilot estimates to choose which treatments to test in the policy learning stage. In this example, the addition of the simulated treatments to the pilot stage leads to the decision maker testing w_2 and w_3 in the policy learning stage and choosing the optimal policy w_3 , in contrast to the RCT approach which would have chosen w_2 .

1.1 An example: Airbnb’s ML experimentation pipeline

Airbnb presents a concrete instance of the funnel above. The company runs thousands of online A/B tests related to ML model product changes (e.g. ranking changes), while the set candidate ideas teams consider (e.g. different models, different implementations etc) awaiting evaluation is on the order of twenty times that number. Figure 3 gives an example of an internal workflow common

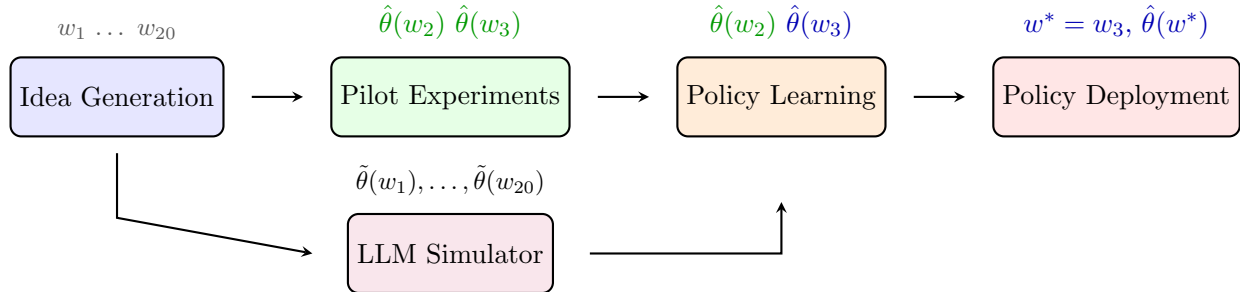


Figure 2: Experimentation funnel with concurrent LLM Simulator: candidate policies w_1 – w_{20} (gray), filtered to w_2, w_3 (green) in pilots, LLM path feeds into policy learning, w_3 (blue) selected for deployment after a large RCT.

for many ML experimentation pipelines: offline development, online experimentation, launch, and post-launch reporting, with feedback to offline modeling at each failure gate. Offline modeling generates candidates far faster than they can be tested online, so experiment capacity is the binding constraint.

Airbnb is aware of this problem and has a prioritization model that relates the rate of welfare growth to the pipeline above. Write $\Delta W \equiv U(\theta_{w^*}) - U(\theta_0)$ for the welfare gain from deploying the best available treatment relative to the status quo, with $w^* \in \arg \max_{w \in \mathcal{W}} U(\theta(w))$ as in the objective above. When bookings are the primary component of U , ΔW coincides with the bookings increment ΔB in Airbnb’s internal accounting. Let T denote models trained, T_s those that pass offline evaluation, E those advanced to online experiments, and E_s those that succeed on business metrics. Then

$$\frac{d}{dt} \Delta W = \underbrace{\min \left(\frac{dT}{dt}, \frac{T}{E} \frac{dE}{dt} \right)}_{\text{experiment capacity}} \times \underbrace{\frac{T_s}{T} \cdot \frac{E_s}{T_s} \cdot \frac{\Delta W_E}{E_s} \cdot \frac{\Delta W}{\Delta W_E}}_{\text{conversion to realized welfare}},$$

where ΔW_E is the in-experiment welfare of successful tests. When offline model production outpaces online testing, $\frac{dT}{dt} > \frac{T}{E} \frac{dE}{dt}$ and the minimum binds on experiment throughput: Airbnb is experiment-limited. Maximizing $\frac{d}{dt} \Delta W$ then requires allocating scarce experiment slots to the candidates with the largest expected $U(\theta_w)$ per unit of experiment cost, which is precisely the screening problem this note formalizes, and one common across any company, lab or research team running large scale experimentation programs.

2 Objective

We adapt the value-of-evidence framework of Abadie et al. (2026), and its empirical-Bayes implementation for pilot selection to the multi-outcome screening problem of Section 1. Empirical Bayes and Bayesian approaches are common approaches in the policy learning literature (Higsbee 2025). The decision maker’s twin tasks are: (i) choose, on the basis of simulator and historical evidence alone, which treatment(s) to advance; and (ii) decide whether to additionally pay for a costly experimental estimate before committing.

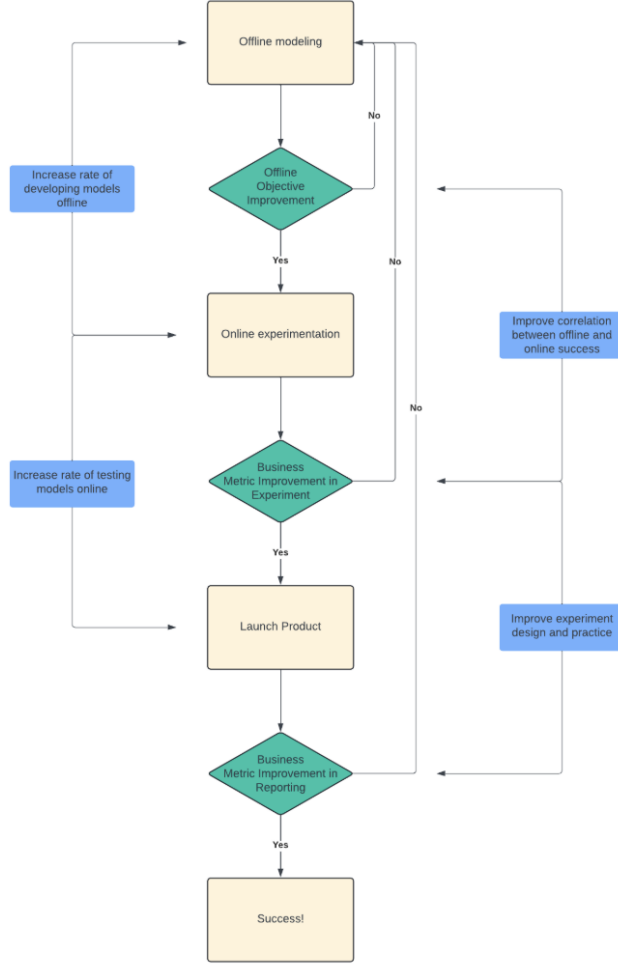


Figure 3: Airbnb’s ML experimentation pipeline: offline development, online experimentation, launch, and reporting, with feedback to offline modeling at each failure gate.

2.1 Decision Problem and Welfare

There is a finite set of candidate treatments

$$\mathcal{W} = \{1, \dots, J\}.$$

Each $w \in \mathcal{W}$ has an unknown vector of true treatment effects

$$\theta_w = (\theta_{w1}, \dots, \theta_{wK}) \in \mathbb{R}^K$$

across K outcomes, possibly partitioned into primary outcomes θ_w^P and guardrail outcomes θ_w^G . Welfare is a scalar function $U : \mathbb{R}^K \rightarrow \mathbb{R}$, and the welfare delivered by treatment w is $U(\theta_w)$.

If θ_w were known, the oracle deployment rule would be

$$w^* \in \arg \max_{w \in \mathcal{W}} U(\theta_w),$$

and, in the binary advance/stop variant relevant to pilot selection, the decision maker would advance treatment w whenever $U(\theta_w) > c_w$, where c_w collects implementation cost, opportunity cost, and risk adjustments associated with treatment w . The realized payoff from advancing w is

$$\Pi(w) = U(\theta_w) - c_w,$$

and the payoff from not advancing is normalized to zero.

Two frictions make this objective hard. First, θ_w is unobserved at decision time and must be estimated from a mixture of simulator output, proposal features, and historical RCTs. Second, experimentation capacity is finite: only K pilots may be run, or a total experimentation budget B may be available. The framework below treats the simulator and the historical data as the inputs to a calibrated posterior over θ_w , and defines a value-of-evidence (VOE) object that ranks pilots by their incremental contribution to the welfare-maximizing decision.

2.2 Welfare

The decision-relevant welfare quantity, prior to running any new pilot, is the calibrated expected welfare

$$\bar{W}(w) = \mathbb{E}[U(\theta_w) \mid \tilde{\theta}_w, X_w, \mathcal{D}_H],$$

where $\tilde{\theta}_w$ is the simulator output, X_w is the proposal feature vector, and $\mathcal{D}_H$ is the body of historical experiments described in Section 2.3. The continuation value of advancing w on the basis of simulator-and-history evidence is

$$V_w^S = \max\{\bar{W}(w) - c_w, 0\},$$

with corresponding action $a_w^S = \mathbf{1}\{\bar{W}(w) > c_w\}$.

A robust variant replaces the point belief by an ambiguity set $\bar{W}(w) \in [L_w, U_w]$ and prescribes

$$\text{advance if } L_w > c_w, \quad \text{stop if } U_w < c_w, \quad \text{pilot if } L_w \leq c_w \leq U_w.$$

The upper and lower bounds U and L can be chosen following Manski bounds (Manski 2003, 2005) given additional assumptions. In the screening variant of Appendix A, in which only M treatments may be A/B-tested and the best of the tested ones is then deployed, the welfare object is the corresponding maximum, $\mathbb{E}[\max_{w \in S} U(\theta_w) \mid \cdot]$. Both variants share the same primitive $\bar{W}(w)$.

2.3 Data: Simulator, Pilot, and Historical Evidence

Three sources of evidence enter the decision.

Simulator output. For each $w \in \mathcal{W}$, before any new pilot is run, the firm observes a (potentially biased and noisy) simulator prediction

$$\tilde{\theta}_w = \theta_w + b_w + \nu_w,$$

where b_w is simulator bias and ν_w is simulation noise. Both can vary with metric, market segment, treatment type, and horizon.

Proposal features. For each candidate w , X_w is a vector of features available before the pilot decision: proposal text, treatment description, target population, business area and product surface, intended primary and guardrail metrics, pre-experiment baseline metrics, prior evidence, and constraints (engineering cost, traffic availability, experimental dependencies). X_w is the information set on which the simulator and the calibration step are allowed to condition.

Pilot estimate. If treatment w is run as a pilot or A/B test of size n_w , the firm eventually observes

$$\hat{\theta}_w = \theta_w + \varepsilon_w, \quad \varepsilon_w \mid \theta_w \sim \mathcal{N}(0, \Sigma_w(n_w)),$$

an unbiased experimental estimate whose covariance $\Sigma_w(n_w)$ depends on sample size, duration, outcome noise, and design.

Historical data. The firm has access to a panel of historical experiments

$$\mathcal{D}_H = \{(\tilde{\theta}_h, \hat{\theta}_h, \Sigma_h, X_h, D_h)\}_{h \in \mathcal{H}},$$

where $\tilde{\theta}_h$ is the temporally-honest simulator prediction available before historical experiment h was run, $\hat{\theta}_h$ is the realized RCT estimate with sampling covariance Σ_h , X_h is the historical feature vector, and D_h records the historical funnel decision (proposed, piloted, escalated, launched, rolled back, failed). The role of $\mathcal{D}_H$ is not to validate whether the simulator is accurate; it is to learn the mapping

$$(\tilde{\theta}, X) \longmapsto \mathcal{L}(\theta \mid \tilde{\theta}, X, \mathcal{D}_H),$$

i.e. the calibrated belief about the true treatment-effect vector that a simulator output and a feature vector imply.

Calibrated belief. For a new candidate w , the calibrated posterior distribution of θ_w is

$$\theta_w \mid \tilde{\theta}_w, X_w, \mathcal{D}_H \sim F_w,$$

with mean and covariance

$$\mu_w = \mathbb{E}[\theta_w \mid \tilde{\theta}_w, X_w, \mathcal{D}_H], \quad V_w = \text{Var}(\theta_w \mid \tilde{\theta}_w, X_w, \mathcal{D}_H).$$

The pair (μ_w, V_w) is the sufficient object for screening. *The simulator output itself is not the decision-relevant evidence; the historically calibrated belief that the simulator induces is.*

Definition 1 (Calibration). *The simulator is calibrated for outcome k if*

$$\mathbb{E}[\theta_{wk} \mid \tilde{\theta}_{wk} = s, X_w, \mathcal{D}_H] = s$$

for all relevant values s . Equivalently, $\mathbb{E}[\theta_{wk} \mid \tilde{\theta}_w, X_w, \mathcal{D}_H] = \tilde{\theta}_{wk}$. When the raw simulator is not perfectly calibrated, the calibrated prediction is

$$\mu_{wk} = \mathbb{E}[\theta_{wk} \mid \tilde{\theta}_w, X_w, \mathcal{D}_H].$$

Calibration is not the same as perfect prediction: it requires the simulator to be correct on average conditional on its forecast and on the proposal features, while perfect prediction would require $\theta_w = \tilde{\theta}_w$ or $V_w = 0$.

2.4 Value of Evidence: Should Treatment w be Piloted?

Before pilot w is run, the experimental estimate $\hat{\theta}_w$ is unobserved. The value of running the pilot is therefore a pre-posterior object: integrate over the predictive distribution of $\hat{\theta}_w$ implied by the calibrated belief F_w and the sampling model $\hat{\theta}_w \mid \theta_w \sim \mathcal{N}(\theta_w, \Sigma_w(n_w))$. After observing $\hat{\theta}_w$, the calibrated expected welfare *post-pilot* is

$$\bar{W}(w \mid \hat{\theta}_w) = \mathbb{E}[U(\theta_w) \mid \tilde{\theta}_w, X_w, \mathcal{D}_H, \hat{\theta}_w].$$

The conditional value of evidence from running pilot w at design n is then

$$\boxed{\text{VOE}_w(n) = \mathbb{E}_{\hat{\theta}_w | \tilde{\theta}_w, X_w, \mathcal{D}_H} \left[\max\{\bar{W}(w | \hat{\theta}_w) - c_w, 0\} \right] - V_w^S - C_w(n),} \quad (1)$$

where $C_w(n)$ is the cost of the pilot at design n (sample size, traffic, engineering, review, calendar time). Equation (1) is the value-of-evidence object adapted to a setting in which the firm already conditions on simulator-and-history evidence: the first term is the expected post-pilot continuation value, V_w^S is the without-pilot continuation value, and $C_w(n)$ is the price paid for the experimental signal.

The pilot rule is to run treatment w whenever $\text{VOE}_w(n) > 0$. Under capacity- K pilot constraints,

$$E^* \in \arg \max_{E \subseteq \mathcal{W}, |E| \leq K} \sum_{w \in E} \max_n \text{VOE}_w(n);$$

under a total experimentation budget B ,

$$(E^*, n^*) \in \arg \max_{E, n} \sum_{w \in E} \text{VOE}_w(n_w) \quad \text{s.t.} \quad \sum_{w \in E} C_w(n_w) \leq B.$$

The decision-theoretic principle is therefore not to pilot the treatments with the largest raw simulated welfare $U(\theta_w)$ or the largest calibrated welfare $\bar{W}(w)$, but to pilot the treatments whose pilot estimate is most likely to change the deployment decision, net of cost.

Closed-form Gaussian special case. Equation (1) admits a closed form when the local-linear approximation of Appendix B applies and the calibrated posterior of θ_w is Gaussian. Project θ_w onto the welfare axis using $a = \nabla U(\theta_0)$:

$$\begin{aligned} \bar{W}(w) &\approx U(\theta_0) + a'(\mu_w - \theta_0), & a'\theta_w | \tilde{\theta}_w, X_w, \mathcal{D}_H &\sim \mathcal{N}(a'\mu_w, \sigma_w^2), & \sigma_w^2 &= a'V_w a, \\ a'\hat{\theta}_w | a'\theta_w &\sim \mathcal{N}(a'\theta_w, \sigma_{\bar{W},w}^2(n)), & \sigma_{\bar{W},w}^2(n) &\equiv a'\Sigma_w(n) a. \end{aligned}$$

Two scalars summarize the entire problem: the calibrated welfare uncertainty σ_w^2 and the welfare-axis pilot variance $\sigma_{\bar{W},w}^2(n)$. Conjugate Gaussian updating gives the post-pilot posterior mean

$$\mu_w^{S,\hat{\theta}} = \lambda_w(n) a'\hat{\theta}_w + (1 - \lambda_w(n)) a'\mu_w, \quad \lambda_w(n) = \frac{\sigma_w^2}{\sigma_w^2 + \sigma_{\bar{W},w}^2(n)} \in [0, 1],$$

where $\lambda_w(n)$ is the shrinkage weight on the pilot. Before observing $\hat{\theta}_w$, the predictive distribution of the post-pilot welfare mean is

$$\bar{W}(w | \hat{\theta}_w) \Big| \tilde{\theta}_w, X_w, \mathcal{D}_H \sim \mathcal{N}(\bar{W}(w), \omega_w^2(n)), \quad \omega_w^2(n) = \lambda_w(n) \sigma_w^2 = \frac{\sigma_w^4}{\sigma_w^2 + \sigma_{\bar{W},w}^2(n)}.$$

Substituting into (1) yields the truncated-normal expression

$$\text{VOE}_w(n) = (\bar{W}(w) - c_w) \Phi(z_w) + \omega_w(n) \phi(z_w) - \max\{\bar{W}(w) - c_w, 0\} - C_w(n), \quad z_w = \frac{\bar{W}(w) - c_w}{\omega_w(n)}, \quad (2)$$

where Φ, ϕ are the standard normal cdf and pdf. Equation (2) is the Gaussian VOE form, expressed in the local-welfare metric $a'\theta$ rather than in the scalar τ_j of the one-outcome benchmark.

Pilot rule and comparative statics. The first three terms in (2) aggregate to the *gross* value of pilot information,

$$G_w(n) \equiv \omega_w(n) \phi(z_w) - [\max\{\bar{W}(w) - c_w, 0\} - (\bar{W}(w) - c_w) \Phi(z_w)],$$

and the rule is to pilot if $G_w(n) > C_w(n)$ at some feasible design n . The bracketed term is the *loss from acting at the prior mean*: if $\bar{W}(w) > c_w$ the firm currently advances and the loss is $(\bar{W}(w) - c_w)[1 - \Phi(z_w)]$; if $\bar{W}(w) < c_w$ the firm currently stops and the loss is $-(\bar{W}(w) - c_w)\Phi(z_w)$. Standard truncated-normal algebra gives

$$G_w(n) = \omega_w(n) [\phi(z_w) - |z_w| (1 - \Phi(|z_w|))].$$

This factorization makes four comparative statics transparent:

- (i) *Threshold distance.* G_w is decreasing in $|z_w|$ and maximized at $\bar{W}(w) = c_w$. A treatment whose calibrated welfare lies far above or far below the threshold is rarely worth piloting; a marginal treatment is.
- (ii) *Calibration uncertainty.* For fixed $|z_w|$, G_w is increasing in $\omega_w(n)$, hence in $\sigma_w^2 = a'V_w a$. Treatments on which the simulator-and-history calibration is informative (σ_w^2 small) are worth less to pilot, holding distance to the threshold fixed.
- (iii) *Pilot precision.* $\omega_w^2(n)$ is decreasing in $\sigma_{\bar{W},w}^2(n)$; a more precise pilot raises G_w at a rate $\partial \omega_w^2 / \partial \sigma_{\bar{W},w}^2 = -\omega_w^4 / \sigma_w^4$. The marginal value of precision is itself proportional to ω_w^4 , so it is largest precisely when the pilot was already going to be informative.
- (iv) *Pilot cost.* The pilot rule is $G_w(n) > C_w(n)$, so a uniformly cheaper pilot expands the set of treatments piloted; an expensive pilot is justified only on high-stakes treatments where $|\bar{W}(w) - c_w|$ and σ_w^2 are both moderate.

Combining (i)–(iv), the optimal pilot *design* solves the first-order condition

$$\phi(z_w^*) \frac{\partial \omega_w}{\partial n} (\bar{W}(w) - c_w) / \omega_w(n^*) = C'_w(n^*),$$

which simplifies, when $\bar{W}(w) \neq c_w$ is held fixed, to $\phi(z_w^*) \partial_n \omega_w(n^*) = C'_w(n^*)$: the firm chooses pilot intensity until the marginal welfare-axis precision delivered by the pilot equals its marginal cost. With a budget B across treatments, the dual problem allocates the pilot budget so that the ratio $\partial_n \text{VOE}_w(n) / \partial_n C_w(n)$ is equalized across w .

Shape of U : primary versus guardrail outcomes. The two scalars σ_w^2 and $\sigma_{\bar{W},w}^2(n)$ inherit the shape of U through a . With the partition $\theta_w = (\theta_w^P, \theta_w^G)$ and $a = (a_P, a_G)$,

$$\begin{aligned} \sigma_w^2 &= a_P^2 V_{w,PP} + 2 a_P a_G V_{w,PG} + a_G^2 V_{w,GG}, \\ \sigma_{\bar{W},w}^2(n) &= a_P^2 \Sigma_{w,PP}(n) + 2 a_P a_G \Sigma_{w,PG}(n) + a_G^2 \Sigma_{w,GG}(n), \end{aligned}$$

so the contribution of outcome k to VOE enters *quadratically through its welfare weight* a_k . Two regimes follow.

- (1) *Guardrail welfare-irrelevant* ($a_G = 0$). Both σ_w^2 and $\sigma_{\bar{W},w}^2(n)$ collapse to $a_P^2 V_{w,PP}$ and $a_P^2 \Sigma_{w,PP}(n)$. Calibration $V_{w,GG}$ on the guardrail axis is irrelevant for $\text{VOE}_w(n)$, and so is pilot precision $\Sigma_{w,GG}(n)$ on the guardrail. Pilot effort should be allocated only to reduce primary-outcome variance.

- (2) *Guardrail binding* ($a_G \neq 0$). For additive welfare $U(\theta) = B(\theta^P) - \sum_g \lambda_g h_g(\theta_g^G)$, $a_G = -\lambda_g h'_g(\theta_{0,g}^G)$; for constrained welfare, a_G is the shadow value $-\lambda_g \partial c_g / \partial \theta_g^G$ (Appendix E). In either case, both calibration and pilot precision on the guardrail axis matter for VOE in proportion to a_G^2 . A treatment whose simulated guardrail effect $\tilde{\theta}_w^G$ pushes the calibrated welfare $\bar{W}(w)$ close to c_w is exactly the case where the closed form predicts piloting: the marginal welfare a_G is large, the calibrated guardrail uncertainty $V_{w,GG}$ is sizeable, and the pilot can resolve it.

A useful corollary follows directly. Suppose the simulator is much better calibrated on guardrails than on primary outcomes, so that $V_{w,GG} \ll V_{w,PP}$, while pilot variances are comparable. Then $\sigma_w^2 \approx a_P^2 V_{w,PP}$ and the welfare-axis pilot variance is dominated by the primary-pilot term $a_P^2 \Sigma_{w,PP}(n)$. Equation (2) therefore prescribes piloting in order to reduce primary-outcome uncertainty, while the calibrated simulator carries the guardrail screening: this is the precise quantitative statement of the rule in Appendix E that simulators should serve as feasibility filters on guardrails and A/B tests should be allocated to learn primary value. The same identity, read in the opposite direction, says that whenever the guardrail simulator is the comparative weakness ($V_{w,GG} \gg V_{w,PP}$ and $a_G \neq 0$), the marginal pilot value scales with $a_G^2 V_{w,GG}$ and pilot capacity should flow to guardrail-resolving experiments.

2.5 Role of Historical Experiments and New Treatments

The historical data $\mathcal{D}_H$ play four distinct roles in the framework above.

Calibration of the simulator. For each historical experiment h , the panel

$$(\tilde{\theta}_h, \hat{\theta}_h, \Sigma_h, X_h, D_h)$$

identifies a conditional model

$$\theta_h = m(\tilde{\theta}_h, X_h) + \eta_h, \quad \hat{\theta}_h = m(\tilde{\theta}_h, X_h) + \eta_h + \varepsilon_h,$$

with $\eta_h \sim F_\eta$ summarizing residual cross-treatment heterogeneity and $\varepsilon_h \mid \theta_h \sim \mathcal{N}(0, \Sigma_h)$ summarizing pilot sampling noise. The calibration map $m(\cdot, \cdot)$ and the residual law F_η are estimated by empirical Bayes, accounting for the sampling variance Σ_h of $\hat{\theta}_h$ so that pilot noise is not absorbed into the calibration. Implementations range from a flexible regression of $\hat{\theta}_h$ on $(\tilde{\theta}_h, X_h)$ with a measurement-error correction, to hierarchical empirical-Bayes shrinkage across product areas, to Gaussian-process or sieve estimators on X .

Temporal honesty. Historical simulator outputs $\tilde{\theta}_h$ must be regenerated using only information available before experiment h was run; otherwise post-treatment information leaks into the calibration and biases μ_w . Out-of-sample validation by time, business area, or treatment family is the practical safeguard.

Funnel decisions. The variable D_h — the historical funnel decision (stopped, piloted, escalated, launched, rolled back, failed) — is informative even when an RCT estimate $\hat{\theta}_h$ is not available. It identifies which proposals were considered comparable in X_h , anchors the selection mix in the historical pilot panel, and provides the comparison against which a simulator-assisted screening rule can be evaluated.

New treatments via historical features. A central practical concern is that a new candidate w may differ from any single historical experiment h . The framework handles this through X_w . The calibrated belief for w is the posterior of $m(\tilde{\theta}, X) + \eta$ at $(\tilde{\theta}_w, X_w)$, which extrapolates by similarity in feature space rather than by exact match. As a transparent illustration, with a similarity kernel $K_h(X_w)$ on the feature space, the local empirical-Bayes posterior mean takes the form

$$\mu_w = \frac{\sum_{h \in \mathcal{H}} K_h(X_w) [\hat{\theta}_h - \tilde{\theta}_h]}{\sum_{h \in \mathcal{H}} K_h(X_w)} + \tilde{\theta}_w,$$

i.e. the simulator output corrected by a feature-weighted average of historical simulator residuals; more general implementations use the ridge or Gaussian-process posterior of $m(\tilde{\theta}, X)$ at $(\tilde{\theta}_w, X_w)$. Three implications follow.

- (i) *Coverage.* The variance V_w widens automatically when X_w lies far from the support of $\{X_h\}_{h \in \mathcal{H}}$. The $\text{VOE}_w(n)$ object then mechanically rises (calibration uncertainty $a'V_w a$ is large), so the framework prescribes piloting precisely the treatments for which historical experiments are uninformative.
- (ii) *Transfer.* A new treatment whose features X_w are similar to past experiments inherits a tight V_w even if its specific simulator prediction $\tilde{\theta}_w$ has never been seen before. This is the precise sense in which historical experiments make simulators useful for new ideas: they license a calibrated belief on θ_w via similarity in X_w , not via memorization of past $\tilde{\theta}$.
- (iii) *Targeted historical investment.* Because V_w depends on the local density of X_h , the firm can plan future RCTs to fill gaps in feature space, expanding the population of X_w for which the simulator can be screened against $\mathcal{D}_H$ without a pilot.

The four roles together — calibration, temporal honesty, funnel-decision evidence, and feature-based extrapolation — are what convert the simulator from a heuristic into the decision-relevant input of equation (1).

3 Building Simulators and the Simulation Brief

Section 2 defines the screening objects (μ_w, V_w) and $\text{VOE}_w(n)$. To produce them in practice, three operational artifacts must be in place: a historical experiment panel $\mathcal{D}_H$, a temporally-honest simulator pipeline that emits $\tilde{\theta}_w$ for any new candidate w , and a calibration model $\mathcal{L}(\theta \mid \tilde{\theta}, X, \mathcal{D}_H)$ that joins the two. This section describes how to build them at Airbnb and what the resulting brief should report to the decision maker.

3.1 Practical Workflow

The retrospective and prospective procedure has eight steps.

1. *Build the historical panel.* For every past proposal h — including those not piloted, when records permit — assemble X_h (proposal text, treatment description, target population, business area and product surface, intended primary and guardrail metrics, pre-experiment baselines, prior evidence, capacity and review constraints), the realized RCT estimate $\hat{\theta}_h$ with sampling covariance Σ_h , and the funnel decision D_h (proposed, piloted, escalated, launched, rolled back, failed).

2. *Generate temporally-honest simulator predictions.* For each historical proposal, run the simulator using only information available before experiment h was launched. Store the raw prediction $\tilde{\theta}_h$ for every metric, the simulation standard error from repeated runs, the prompt and model version, the synthetic-population or persona configuration, segment-level outputs, and any mechanism explanations the simulator emits. Temporal honesty is not optional: if outcomes from h leak into the prompt, the calibration map is biased.
3. *Estimate calibration.* Fit the empirical-Bayes model

$$\hat{\theta}_h = m(\tilde{\theta}_h, X_h) + \eta_h + \varepsilon_h, \quad \eta_h \sim F_\eta, \quad \varepsilon_h \mid \theta_h \sim \mathcal{N}(0, \Sigma_h),$$

with a measurement-error correction so that pilot sampling noise is not absorbed into F_η . Implementations range from a flexible regression of $\hat{\theta}_h$ on $(\tilde{\theta}_h, X_h)$ to hierarchical empirical-Bayes shrinkage by product area, to Gaussian-process or sieve estimators on X .

4. *Validate out of sample.* Hold out historical experiments by time, business area, or treatment family. Evaluate coverage of V_w , sign accuracy, rank accuracy, and *decision* accuracy — the fraction of correct advance/stop calls a calibrated rule would have made on past proposals.
5. *Compute the calibrated posterior for new candidates.* For each new idea w , produce μ_w and V_w , and the welfare-axis projections $\bar{W}(w)$ and $\sigma_w^2 = a'V_w a$.
6. *Compute pilot designs and VOE.* For each candidate w and each feasible pilot size or duration n , evaluate the pilot welfare-variance $\sigma_{\bar{W},w}^2(n) = a'\Sigma_w(n)a$, the predictive variance $\omega_w^2(n)$, and $\text{VOE}_w(n)$ from (2).
7. *Rank and recommend.* Rank candidates by $\max_n \text{VOE}_w(n)$ subject to capacity K or budget B . Report the recommended action: *stop*, *revise*, *pilot at n^** , *scale*, or *launch*.
8. *Re-train the calibration map.* As new pilots realize, append $(\tilde{\theta}_w, \hat{\theta}_w, \Sigma_w, X_w, D_w)$ to $\mathcal{D}_H$ and re-estimate $m(\cdot, \cdot)$ and F_η on a regular cadence; this is what closes the loop between simulator engineering and pilot selection.

The two artifacts that distinguish a useful Airbnb implementation from a heuristic one are temporal honesty in step 2 and the measurement-error correction in step 3. Without temporal honesty, calibration overstates the simulator; without the measurement-error correction, $\text{Var}(\eta_h)$ confounds true heterogeneity with pilot noise and V_w is biased downward.

3.2 The Simulation Brief

For each new proposal w the simulation brief should report, in the following order:

1. Raw simulator prediction $\tilde{\theta}_w$ for each metric, with simulation standard errors from repeated simulator runs.
2. Calibrated posterior mean μ_w and covariance V_w , summarized on the welfare axis by $\bar{W}(w)$ and $\sigma_w^2 = a'V_w a$.
3. Calibrated welfare interval $[L_w, U_w]$ and the probability $\Pr(\bar{W}(w) > c_w \mid \tilde{\theta}_w, X_w, \mathcal{D}_H)$ of exceeding the advancement threshold.
4. Historical calibration diagnostics for the closest comparator experiments: effective sample size in feature space, residual scale, sign-accuracy, decision-accuracy.

5. Expected pilot welfare-variance $\sigma_{\tilde{W},w}^2(n) = a' \Sigma_w(n) a$ at each feasible pilot size or duration n , broken down per metric.
6. Conditional value of evidence $\text{VOE}_w(n)$ at each feasible n , with the optimal pilot intensity $n^* = \arg \max_n \text{VOE}_w(n)$.
7. Per-axis decomposition of σ_w^2 and $\sigma_{\tilde{W},w}^2(n^*)$ into primary and guardrail contributions, so the decision maker sees which metric the pilot will resolve.
8. Recommended action and rationale: *stop*, *revise*, *pilot at n^** , *scale*, or *launch*.

Items 2–7 collectively make the simulator’s claim falsifiable: every recommendation is tied to a calibrated posterior, a calibration diagnostic on similar past experiments, and an explicit per-metric reduction in welfare variance the pilot is expected to deliver.

3.3 Example: Reducing the Cancellation Window

Suppose Airbnb is considering shortening the free-cancellation window from 48 to 24 hours on a particular booking surface. Two outcomes enter welfare: bookings θ_w^P (primary; positive direction) and host-cancellation rate θ_w^G (guardrail; negative direction). The decision maker uses the locally-linear welfare $U(\theta) = \theta^P - 2\theta^G$, with advancement threshold $c_w = 0.20$ welfare units (the implementation, review, and rollout cost of the change).

Calibrated belief. The simulator returns

$$\tilde{\theta}_w = (\tilde{\theta}_w^P, \tilde{\theta}_w^G) = (1.0, 0.5) \quad (\text{percent change in each outcome}).$$

The historical empirical-Bayes calibration, fitted on past cancellation-policy and platform-friction experiments at Airbnb, has $R_P^2 = 0.55$ on bookings and $R_G^2 = 0.80$ on host cancellations, giving

$$\mu_w = (0.55 \cdot 1.0, 0.80 \cdot 0.5) = (0.55, 0.40), \quad V_w = \text{diag}(0.36, 0.05).$$

On the welfare axis,

$$\bar{W}(w) = 0.55 - 2 \cdot 0.40 = -0.25, \quad \sigma_w^2 = 1^2 \cdot 0.36 + (-2)^2 \cdot 0.05 = 0.56, \quad \sigma_w \approx 0.75.$$

Because $\bar{W}(w) < c_w$, the without-pilot continuation value is $V_w^S = 0$: the firm currently *stops*. But the calibration uncertainty is large, $\sigma_w \approx 0.75$, and the threshold distance is only $|\bar{W}(w) - c_w|/\sigma_w \approx 0.6$, so a pilot can plausibly flip the decision.

Pilot designs. Three feasible designs and their VOE summaries are reported in the brief as Table 1. The standard pilot n_2 delivers the largest net VOE: gross VOE rises with n but is concave, while costs are roughly convex, so the optimum is interior.

Per-axis decomposition. The pre-pilot welfare uncertainty decomposes as

$$\sigma_w^2 = \underbrace{a_P^2 V_{w,PP}}_{=0.36} + \underbrace{a_G^2 V_{w,GG}}_{=0.20} = 0.56,$$

so the primary axis carries 64% of welfare uncertainty even though the welfare weight $|a_G| = 2$ is twice a_P , because the well-calibrated guardrail simulator ($R_G^2 = 0.80$) has already shrunk $V_{w,GG}$. Most of the standard pilot’s welfare value therefore comes from reducing primary-bookings uncertainty, not guardrail uncertainty: this example is the empirical instance of the rule in Appendix E, in which the simulator carries the guardrail and the pilot resolves the primary.

Design	$\sigma_{\bar{W},w}^2(n)$	$\omega_w^2(n)$	z_w	gross VOE	cost $C_w(n)$	net VOE
Small pilot (n_1)	0.50	0.30	-0.83	0.06	0.02	0.04
Standard pilot (n_2)	0.15	0.44	-0.68	0.10	0.05	0.05
Large pilot (n_3)	0.05	0.51	-0.63	0.12	0.20	-0.08

Table 1: Simulation brief, cancellation-window example. Gross VOE is $(\bar{W}(w) - c_w)\Phi(z_w) + \omega_w(n)\phi(z_w) - \max\{\bar{W}(w) - c_w, 0\}$; net VOE subtracts the pilot cost.

Recommendation. The brief recommends *pilot at n_2* , with predictive distribution of the post-pilot welfare mean

$$\bar{W}(w \mid \hat{\theta}_w) \mid \tilde{\theta}_w, X_w, \mathcal{D}_H \sim \mathcal{N}(-0.25, 0.44).$$

If, after observing $\hat{\theta}_w$, the post-pilot mean $\mu_w^{S,\hat{\theta}}$ exceeds $c_w = 0.20$, advance the change; otherwise stop. The simulator brief is therefore not a single point estimate, but an end-to-end recipe that maps simulator and historical evidence into a calibrated, falsifiable decision.

3.4 Example: Allocating One Pilot Slot Across Multiple Treatments

The previous example takes a single proposal and asks whether it should be piloted. The harder, more typical Airbnb problem is the dual one: a portfolio of candidate treatments competes for a single pilot slot, and the firm must decide which candidate to send to the pilot and which to either launch or stop on the basis of simulator-and-history evidence alone. The VOE framework gives a precise rule.

Suppose three candidate treatments $\{w_1, w_2, w_3\}$ are under review on the same booking surface, all evaluated against the welfare $U(\theta) = \theta^P - 2\theta^G$ with advancement threshold $c = 0.20$:

- w_1 : a redesigned search-ranking model (a class of treatment for which Airbnb has run many past A/B tests).
- w_2 : a new host-onboarding flow with mentor matching (small set of historical comparators on the host side).
- w_3 : a simplified cancellation policy aimed at a niche cohort (close to past cancellation experiments, but the cohort is small and the predicted effect is negative).

The simulator and the calibration map of Section 3.1 produce a brief entry per candidate. To keep the comparison clean, suppose each candidate would be piloted at the same standard size, with $\sigma_{\bar{W},w_j}^2(n) = 0.10$.

Cand.	$\bar{W}(w_j)$	$\sigma_{w_j}^2$	$\omega_{w_j}^2(n)$	z_{w_j}	gross VOE	cost	action
w_1 (search)	0.80	0.09	0.04	2.90	0.001	0.05	launch
w_2 (onboarding)	0.25	0.25	0.18	0.12	0.144	0.05	pilot
w_3 (cancellation)	-0.40	0.04	0.01	-5.61	0.000	0.05	stop

Table 2: Multi-treatment simulation brief. Gross VOE is computed from (2) with $c = 0.20$; net VOE is gross minus cost. Only w_2 has positive net VOE.

Reading the brief. Without the simulator, all three candidates would compete for the single pilot slot on the basis of stakeholder priors, team conviction, or raw simulator output (which would rank w_1 first, w_2 second, w_3 last). The calibrated VOE in Table 2 gives a different and sharper allocation:

- w_1 has $\bar{W}(w_1) = 0.80$ and $\sigma_{w_1} = 0.30$, so the threshold distance $|\bar{W}(w_1) - c|/\sigma_{w_1} = 2.0$ is large. Even an informative pilot has essentially no chance of flipping the calibrated belief below c ; gross VOE is 0.001, well below cost. The brief recommends *launch* on the basis of simulator and history alone.
- w_3 has $\bar{W}(w_3) = -0.40$ and $\sigma_{w_3} = 0.20$, with threshold distance 3.0 on the other side. Gross VOE is essentially zero. The brief recommends *stop*; no pilot capacity should be spent.
- w_2 is the marginal candidate. Its calibrated welfare $\bar{W}(w_2) = 0.25$ sits just above threshold, but the calibration uncertainty $\sigma_{w_2} = 0.50$ is large, $z_{w_2} \approx 0.12$ is close to zero, and a pilot of standard size produces $\omega_{w_2}^2(n) = 0.18$ in welfare-axis predictive variance. Gross VOE is 0.144, comfortably above the same cost the other two cannot cover. The brief recommends *pilot*.

The simulator therefore plays *two complementary roles in pilot allocation*, both of which are absent in a heuristic workflow. First, it screens out clearly-launch and clearly-stop candidates (w_1 and w_3), so pilot capacity is not wasted on questions whose calibrated answer is already known. Second, among candidates whose calibrated answer is not yet known, it identifies the one whose pilot is most likely to change the decision and routes capacity to it. The treatment that is ultimately deployed at scale (w_1 now, w_2 pending pilot) is determined by the joint use of simulator-and-history evidence and pilot evidence — not by the pilot alone, and not by the simulator alone.

Sensitivity to capacity. If the firm has two pilot slots rather than one, the brief promotes the candidate with the next-largest net VOE; in this example both w_1 and w_3 have negative net VOE and the second slot is left unused (or carried over to a future cohort). If the firm has zero pilot slots, the simulator brief still produces the deterministic decision rule of advancing every w with $\bar{W}(w) > c$ — here, w_1 and w_2 — but at the cost of acting on w_2 at large calibration uncertainty. Both cases are read off the same brief; this is the operational sense in which the simulator allows the firm to *run the right pilot for deploying the right treatment*.

A Exact Screening Problem

Suppose the firm can A/B test only $M < J$ candidates. A screening rule chooses a subset

$$S \subseteq \mathcal{W}, \quad |S| = M.$$

For simplicity, assume first that the A/B test is perfectly informative: once a treatment is tested, the firm learns its true welfare and then launches the best tested treatment. The ex ante value of screening set S is

$$V(S) = \mathbb{E} \left[\max_{w \in S} U(\theta_w) \mid \{\tilde{\theta}_w\}_{w \in \mathcal{W}}, \mathcal{D}_H \right].$$

Theorem 1 (Exact optimal screening). *Suppose the firm can test M candidates and, after testing, can deploy the best tested treatment. Then the optimal screening set solves*

$$S^* \in \arg \max_{S \subseteq \mathcal{W}, |S|=M} \mathbb{E} \left[\max_{w \in S} U(\theta_w) \mid \{\tilde{\theta}_w\}_{w \in \mathcal{W}}, \mathcal{D}_H \right].$$

If $M = 1$, this reduces to

$$w^* \in \arg \max_{w \in \mathcal{W}} \bar{W}(w),$$

where

$$\bar{W}(w) = \mathbb{E} \left[U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H \right].$$

Proof. If the firm tests set S , it observes the welfare of every candidate $w \in S$. Since it can deploy the best tested candidate, its realized payoff is

$$\max_{w \in S} U(\theta_w).$$

Before testing, the firm does not know the true vectors θ_w , but it knows their posterior distribution conditional on the simulator outputs and the historical calibration data. Hence the expected payoff from testing S is

$$V(S) = \mathbb{E} \left[\max_{w \in S} U(\theta_w) \mid \{\tilde{\theta}_w\}_{w \in \mathcal{W}}, \mathcal{D}_H \right].$$

The optimal set is therefore any size- M set maximizing $V(S)$.

If $M = 1$, then $S = \{w\}$ for some w , and

$$V(\{w\}) = \mathbb{E} \left[U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H \right] = \bar{W}(w).$$

Thus the optimal one-candidate screening rule maximizes $\bar{W}(w)$ over $w \in \mathcal{W}$. $\square$

The theorem says that the correct object for screening is not the raw simulated welfare $U(\tilde{\theta}_w)$, but rather the calibrated expected welfare $\mathbb{E}[U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H]$.

B Taylor Approximation

The expected welfare $\bar{W}(w)$ may be hard to compute when U is nonlinear. A local Taylor approximation gives a transparent screening rule.

Fix a reference point

$$\theta_0 \in \mathbb{R}^K,$$

which may be the status quo, the prior mean, or the average historical treatment effect. Suppose U is twice continuously differentiable near θ_0 . A second-order Taylor expansion gives

$$U(\theta_w) = U(\theta_0) + \nabla U(\theta_0)'(\theta_w - \theta_0) + R_w,$$

where the remainder is

$$R_w = \frac{1}{2}(\theta_w - \theta_0)'H(\bar{\theta}_w)(\theta_w - \theta_0)$$

for some point $\bar{\theta}_w$ between θ_w and θ_0 , and where $H(\cdot)$ is the Hessian matrix of U .

Define the local welfare-gradient vector

$$a = \nabla U(\theta_0) = \left(\frac{\partial U}{\partial \theta_1}(\theta_0), \dots, \frac{\partial U}{\partial \theta_K}(\theta_0) \right)'.$$

Then

$$U(\theta_w) \approx U(\theta_0) + a'(\theta_w - \theta_0).$$

Taking conditional expectations gives

$$\bar{W}(w) = \mathbb{E}[U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H] \approx U(\theta_0) + a'(\mu_w - \theta_0).$$

Since $U(\theta_0)$ and $-a'\theta_0$ are constant across w , ranking candidates by approximate expected welfare is equivalent to ranking them by

$$a'\mu_w.$$

Corollary 1 (Local expected-welfare screening). *Suppose U is twice continuously differentiable in a neighborhood of θ_0 . Let*

$$a = \nabla U(\theta_0).$$

Assume the posterior distribution of each θ_w is concentrated near θ_0 , in the sense that

$$r_w^2 = \mathbb{E} \left[\|\theta_w - \theta_0\|^2 \mid \tilde{\theta}_w, \mathcal{D}_H \right]$$

is small. Then

$$\bar{W}(w) = U(\theta_0) + a'(\mu_w - \theta_0) + O(r_w^2).$$

Therefore, if the firm can test one candidate, the locally optimal screening rule is

$$w_{\text{local}}^* \in \arg \max_{w \in \mathcal{W}} a'\mu_w.$$

If the firm uses the plug-in approximation that ranks candidates by their individual expected welfare scores, then the size- M local screening set is

$$S_{\text{local}}^* = \text{TopM}_{w \in \mathcal{W}} \{a'\mu_w\}.$$

Proof. By Taylor's theorem,

$$U(\theta_w) = U(\theta_0) + a'(\theta_w - \theta_0) + R_w,$$

where

$$R_w = \frac{1}{2}(\theta_w - \theta_0)'H(\bar{\theta}_w)(\theta_w - \theta_0).$$

Taking conditional expectations yields

$$\bar{W}(w) = U(\theta_0) + a' \mathbb{E}[\theta_w - \theta_0 \mid \tilde{\theta}_w, \mathcal{D}_H] + \mathbb{E}[R_w \mid \tilde{\theta}_w, \mathcal{D}_H].$$

By definition,

$$\mathbb{E}[\theta_w \mid \tilde{\theta}_w, \mathcal{D}_H] = \mu_w,$$

so

$$\bar{W}(w) = U(\theta_0) + a'(\mu_w - \theta_0) + \mathbb{E}[R_w \mid \tilde{\theta}_w, \mathcal{D}_H].$$

If the Hessian is bounded near θ_0 , then for some constant C ,

$$|\mathbb{E}[R_w \mid \tilde{\theta}_w, \mathcal{D}_H]| \leq C \mathbb{E}[\|\theta_w - \theta_0\|^2 \mid \tilde{\theta}_w, \mathcal{D}_H] = O(r_w^2).$$

Hence

$$\bar{W}(w) = U(\theta_0) + a'(\mu_w - \theta_0) + O(r_w^2).$$

For $M = 1$, the exact optimal rule maximizes $\bar{W}(w)$. Dropping the second-order term gives the local rule

$$\arg \max_w \{U(\theta_0) + a'(\mu_w - \theta_0)\}.$$

The terms $U(\theta_0)$ and $-a'\theta_0$ are constant across w , so this is equivalent to maximizing $a'\mu_w$. The plug-in size- M rule selects the M largest values of the same score. $\square$

C Risk Adjustment

The local rule above is risk-neutral. If the firm is cautious about simulator uncertainty, it may use a lower-confidence-bound score. Under the local approximation,

$$U(\theta_w) \approx U(\theta_0) + a'(\theta_w - \theta_0),$$

so

$$\text{Var}(U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H) \approx a' V_w a.$$

Corollary 2 (Conservative local screening). *Suppose welfare is locally linear around θ_0 :*

$$U(\theta_w) \approx U(\theta_0) + a'(\theta_w - \theta_0).$$

Suppose the firm evaluates each candidate using the lower confidence bound

$$LCB(w) = \mathbb{E}[U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H] - \rho \sqrt{\text{Var}(U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H)},$$

where $\rho \geq 0$ is a caution parameter. Then, up to the local approximation,

$$LCB(w) = \text{constant} + a'\mu_w - \rho \sqrt{a' V_w a}.$$

Thus the conservative local screening rule is

$$w_{\text{LCB}}^* \in \arg \max_{w \in \mathcal{W}} \left\{ a'\mu_w - \rho \sqrt{a' V_w a} \right\}.$$

Proof. Under the local approximation,

$$U(\theta_w) \approx U(\theta_0) + a'(\theta_w - \theta_0).$$

Taking conditional expectations gives

$$\mathbb{E}[U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H] \approx U(\theta_0) + a'(\mu_w - \theta_0).$$

Taking conditional variances gives

$$\text{Var}(U(\theta_w) \mid \tilde{\theta}_w, \mathcal{D}_H) \approx \text{Var}(a'\theta_w \mid \tilde{\theta}_w, \mathcal{D}_H) = a'V_w a.$$

Substituting these two expressions into the definition of $LCB(w)$ yields

$$LCB(w) \approx U(\theta_0) + a'(\mu_w - \theta_0) - \rho\sqrt{a'V_w a}.$$

The terms $U(\theta_0)$ and $-a'\theta_0$ are constant across w , so maximizing $LCB(w)$ is equivalent to maximizing

$$a'\mu_w - \rho\sqrt{a'V_w a}.$$

□

D When Does a Guardrail Metric Matter for Screening?

Let g index a guardrail coordinate. Write μ_{wg} for the calibrated posterior mean of guardrail metric g for treatment w , and write a_g for the g th component of the local welfare-gradient vector a .

The local expected-welfare score is

$$s_w = a'\mu_w.$$

For a fixed guardrail coordinate g , define the score excluding that coordinate by

$$s_w^{-g} = \sum_{k \neq g} a_k \mu_{wk}.$$

Then

$$s_w = s_w^{-g} + a_g \mu_{wg}.$$

Definition 2 (Pivotal guardrail metric). *For a size- M screening rule that selects the M largest values of $s_w = a'\mu_w$, guardrail metric g is pivotal if the selected set changes when the contribution of metric g is removed. That is,*

$$\text{TopM}_{w \in \mathcal{W}}\{s_w\} \neq \text{TopM}_{w \in \mathcal{W}}\{s_w^{-g}\}.$$

Corollary 3 (Necessary and sufficient conditions for a guardrail metric to affect screening). *Consider the local plug-in screening rule*

$$S_M = \text{TopM}_{w \in \mathcal{W}}\{s_w\}, \quad s_w = a'\mu_w.$$

Assume there are no ties at the M th cutoff either with or without coordinate g . Let

$$S_M^{-g} = \text{TopM}_{w \in \mathcal{W}}\{s_w^{-g}\}.$$

Then guardrail metric g is pivotal if and only if there exist candidates w and v such that

$$w \in S_M, \quad v \notin S_M,$$

and

$$s_w^{-g} < s_v^{-g} \quad \text{while} \quad s_w^{-g} + a_g \mu_{wg} > s_v^{-g} + a_g \mu_{vg}.$$

Equivalently, g is pivotal if and only if there exist $w \in S_M$ and $v \notin S_M$ such that

$$a_g(\mu_{wg} - \mu_{vg}) > s_v^{-g} - s_w^{-g}.$$

In particular, a necessary condition for guardrail metric g to be pivotal is

$$a_g \neq 0 \quad \text{and} \quad \mu_{wg} \neq \mu_{vg} \text{ for some } w, v \in \mathcal{W}.$$

That is, the guardrail metric must both matter locally for welfare and vary across candidates after calibration.

A sufficient primitive condition on the welfare function is

$$\frac{\partial U}{\partial \theta_g}(\theta_0) \neq 0,$$

together with calibrated variation in the metric,

$$\mu_{wg} \neq \mu_{vg} \text{ for some } w, v.$$

For additive welfare of the form

$$U(\theta) = B(\theta^P) - \sum_{g \in G} \lambda_g h_g(\theta_g^G),$$

we have

$$a_g = \frac{\partial U}{\partial \theta_g^G}(\theta_0) = -\lambda_g h'_g(\theta_{0g}^G).$$

Thus guardrail metric g can be pivotal only if

$$\lambda_g h'_g(\theta_{0g}^G) \neq 0$$

and its calibrated posterior mean varies across candidate treatments.

For constrained welfare of the form

$$\max_w B(\theta_w^P) \quad \text{subject to} \quad c_g(\theta_w^G) \leq 0,$$

the corresponding Lagrangian is

$$\mathcal{L}(\theta, \lambda) = B(\theta^P) - \sum_{g \in G} \lambda_g c_g(\theta^G), \quad \lambda_g \geq 0.$$

The guardrail metric g enters the local screening score through the shadow value

$$a_g^{\mathcal{L}} = -\lambda_g \frac{\partial c_g}{\partial \theta_g^G}(\theta_0^G).$$

Therefore, in the constrained case, the primitive condition for guardrail metric g to affect screening is that the constraint has positive local shadow value,

$$\lambda_g > 0,$$

that the constraint is locally sensitive to the metric,

$$\frac{\partial c_g}{\partial \theta_g^G}(\theta_0^G) \neq 0,$$

and that the calibrated guardrail prediction varies across treatments.

Proof. The local screening rule selects the M candidates with highest values of

$$s_w = s_w^{-g} + a_g \mu_{wg}.$$

Removing coordinate g gives the rule that selects the M candidates with highest values of s_w^{-g} .

Under the no-ties assumption, the selected set changes if and only if at least one candidate included under the full score is not included under the score without g , and at least one candidate excluded under the full score is included when g is removed. Equivalently, there exist candidates w and v such that w is selected under the full score, v is not selected under the full score, but v outranks w when coordinate g is removed. That is,

$$s_w^{-g} < s_v^{-g}$$

while

$$s_w = s_w^{-g} + a_g \mu_{wg} > s_v^{-g} + a_g \mu_{vg} = s_v.$$

Rearranging the second inequality gives

$$a_g(\mu_{wg} - \mu_{vg}) > s_v^{-g} - s_w^{-g}.$$

This proves the necessary and sufficient condition for pivotality.

If $a_g = 0$, then $s_w = s_w^{-g}$ for every w , so removing coordinate g cannot change the selected set. If μ_{wg} is constant across all w , then $a_g \mu_{wg}$ is the same additive constant for every candidate and cannot change rankings. Hence $a_g \neq 0$ and cross-candidate variation in μ_{wg} are necessary for pivotality.

The primitive condition follows because, by definition,

$$a_g = \frac{\partial U}{\partial \theta_g}(\theta_0).$$

Thus $a_g \neq 0$ exactly means that welfare has nonzero local marginal sensitivity to guardrail metric g .

For additive welfare,

$$U(\theta) = B(\theta^P) - \sum_{g \in G} \lambda_g h_g(\theta_g^G),$$

differentiating with respect to θ_g^G gives

$$a_g = -\lambda_g h'_g(\theta_{0g}^G).$$

Therefore $a_g \neq 0$ if and only if $\lambda_g h'_g(\theta_{0g}^G) \neq 0$.

For constrained welfare, the local welfare-relevant object is the Lagrangian. Differentiating

$$\mathcal{L}(\theta, \lambda) = B(\theta^P) - \sum_{g \in G} \lambda_g c_g(\theta^G)$$

with respect to θ_g^G gives

$$a_g^{\mathcal{L}} = -\lambda_g \frac{\partial c_g}{\partial \theta_g^G}(\theta_0^G).$$

Thus guardrail metric g has nonzero local shadow value precisely when the multiplier is positive, the constraint is locally sensitive to the metric, and the calibrated metric varies across treatments. This completes the proof. $\square$

E Guardrail-Only Simulation

The previous corollary characterizes when a guardrail metric can affect the local ranking. A related question is when simulation should be used only for guardrails.

Suppose the simulator is informative for guardrails but not for primary outcomes:

$$\mathbb{E}[\theta_w^P \mid \tilde{\theta}_w, \mathcal{D}_H] = \mathbb{E}[\theta_w^P \mid \mathcal{D}_H],$$

while

$$\Pr(\theta_w^G \in \mathcal{G} \mid \tilde{\theta}_w^G, \mathcal{D}_H)$$

varies with the simulated guardrail outcomes. If guardrails enter as feasibility constraints, then the simulator should not be used to rank total welfare. It should instead define the feasible testing set

$$\mathcal{W}_G = \left\{ w \in \mathcal{W} : \Pr(\theta_w^G \in \mathcal{G} \mid \tilde{\theta}_w^G, \mathcal{D}_H) \geq 1 - \alpha \right\}.$$

Then A/B tests should be allocated among candidates in $\mathcal{W}_G$ to learn primary effects.

Thus the correct guardrail-only rule is

$$\max_{w \in \mathcal{W}_G} \text{VOI}_P^{AB}(w),$$

not

$$\max_{w \in \mathcal{W}} U(\tilde{\theta}_w^G).$$

F Main Takeaway

The local score is

$$a' \mu_w.$$

The vector $a = \nabla U(\theta_0)$ captures how much each outcome matters for welfare. The vector $\mu_w = \mathbb{E}[\theta_w \mid \tilde{\theta}_w, \mathcal{D}_H]$ captures what the calibrated simulator implies about true effects. Therefore a simulated metric affects screening only when it is both welfare-relevant and calibrated enough to create cross-treatment variation in posterior beliefs.

In short,

$$\text{simulation weight} = \text{welfare sensitivity} \times \text{calibrated informativeness}.$$

If only guardrail outcomes are well calibrated, then simulations should be used as feasibility screens or risk controls, while A/B tests should be used to learn primary value.

G Simulation: Shadow Cost, Calibration, and Treatment Choice

The screening theorem and its local-approximation corollary make a sharp prediction: the only object that matters for ranking treatments is the vector $a'\mu_w$, where $a = \nabla U(\theta_0)$ is the welfare gradient and μ_w is the calibrated posterior mean. This implies that the effective weight a simulator places on outcome k when choosing treatments is the product of the welfare sensitivity a_k and a calibrated-informativeness scalar that lives in $[0, 1]$. The simulation in this section quantifies this prediction. It maps out, as a function of the guardrail shadow cost a and of the calibration quality of each outcome, (i) which screening rule maximizes expected welfare, (ii) where simulator-engineering effort should be directed, and (iii) when a less policy-relevant but well-calibrated metric should be preferred to a highly policy-relevant but poorly-calibrated one.

G.1 Design

Primitive objects. The economy contains J candidate treatments, indexed by $w \in \{1, \dots, J\}$, each with a two-dimensional true effect

$$\theta_w = (\theta_w^P, \theta_w^G) \in \mathbb{R}^2,$$

where P denotes the primary outcome and G a single guardrail outcome. We draw $\theta_w^P \sim \mathcal{N}(0, \sigma_P^2)$ and $\theta_w^G \sim \mathcal{N}(0, \sigma_G^2)$ independently across treatments and outcomes. The cross-treatment standard deviations σ_P, σ_G play the role of a prior dispersion of true effects: in the calibration analysis they generate the cross-section against which the simulator's predictions can be informative or noisy.

Simulator and calibration quality. For each treatment and outcome, the simulator returns

$$\tilde{\theta}_w^k = \theta_w^k + \nu_w^k, \quad \nu_w^k \sim \mathcal{N}(0, \tau_k^2), \quad k \in \{P, G\},$$

with ν_w^k independent across w, k and independent of θ . The historical calibration data $\mathcal{D}_H$ are assumed to identify the noise variance τ_k^2 (in practice, by regressing past A/B-test effects on past simulator predictions). Under the conjugate Gaussian model, the calibrated posterior mean of Definition 1 takes the shrinkage form

$$\mu_w^k = \mathbb{E}[\theta_w^k \mid \tilde{\theta}_w^k, \mathcal{D}_H] = R_k^2 \tilde{\theta}_w^k, \quad R_k^2 = \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} \in [0, 1].$$

The quantity R_k^2 is exactly the population R^2 of the regression of θ_w^k on $\tilde{\theta}_w^k$; it is our scalar summary of *calibrated informativeness* for outcome k . The two extreme cases are useful intuition: $R_k^2 = 1$ corresponds to a perfectly informative (and trivially calibrated) simulator, while $R_k^2 = 0$ corresponds to a simulator whose output contains no information about the true effect, and whose calibrated prediction collapses to the prior mean 0.

Welfare. We use a locally linear welfare of the form

$$U(\theta_w) = \theta_w^P - a \theta_w^G, \quad a \geq 0,$$

which is exactly the additive welfare in Appendix D with $a_P = 1$ and $a_G = -a$. The parameter a is the guardrail shadow cost: a small means the guardrail has little local welfare weight; a large means a one-unit deterioration in the guardrail destroys a units of primary value. In the constrained interpretation of the note, a plays the role of the Lagrange multiplier λ_g on the guardrail constraint.

Screening rules. For a fixed configuration (a, R_P^2, R_G^2) we draw a population of J candidates and a simulator realization, then choose a single treatment w^* under each of the following rules and record the realized welfare $U(\theta_{w^*})$.

- (i) **Oracle.** $w^* \in \arg \max_w U(\theta_w)$. This is the post-experiment ideal benchmark.
- (ii) **Naive.** $w^* \in \arg \max_w \{\tilde{\theta}_w^P - a\tilde{\theta}_w^G\}$. The firm treats the raw simulator output as if it were the truth.
- (iii) **Calibrated.** $w^* \in \arg \max_w \{\mu_w^P - a\mu_w^G\}$. This is the theorem-implied rule: maximize $a'\mu_w$.
- (iv) **Ignore guardrail.** $w^* \in \arg \max_w \mu_w^P$. The firm acts as if $a = 0$ in screening.
- (v) **Guardrail-as-feasibility filter.** $w^* \in \arg \max_w \{\mu_w^P : \mu_w^G \leq q\}$ where q is the median of μ_w^G . This implements the guardrail-only rule of Appendix E: the simulator is used only to define the feasible testing set.
- (vi) **Random.** A uniform random treatment, the lower benchmark.

We compute the expected realized welfare $\mathbb{E}[U(\theta_{w^*})]$ for each rule by Monte Carlo, sweeping over (a, R_P^2, R_G^2) on a grid. To suppress noise in difference surfaces we use common random numbers across the configurations entering any single comparison: the underlying draws of $\{\theta_w\}$ and of the standardized simulator noise are fixed, and changes in (a, R_P^2) or R_G^2 act only by reweighting these draws.

Algorithm 1 One Monte Carlo replication for fixed (a, R_P^2, R_G^2) .

- 1: Draw $\theta_w^P \sim \mathcal{N}(0, \sigma_P^2)$ and $\theta_w^G \sim \mathcal{N}(0, \sigma_G^2)$ for $w = 1, \dots, J$.
 - 2: Draw standardized noise $z_w^k \sim \mathcal{N}(0, 1)$ for w, k , and set $\tilde{\theta}_w^k = \theta_w^k + \tau_k z_w^k$ with $\tau_k = \sigma_k \sqrt{1/R_k^2 - 1}$.
 - 3: Form calibrated means $\mu_w^k = R_k^2 \tilde{\theta}_w^k$.
 - 4: Compute true welfare $U(\theta_w) = \theta_w^P - a\theta_w^G$.
 - 5: For each rule, pick w^* from its score and record $U(\theta_{w^*})$.
-

We use $J = 50$ candidate treatments and 8,000 Monte Carlo replications per grid point (and 25,000 for figure 5, which differences two close quantities); $\sigma_P = \sigma_G = 1$ throughout. Full source code is provided in `simulation/run_simulation.py`.

G.2 Result 1: when does calibrating the guardrail matter?

Figure 4 fixes the primary calibration at $R_P^2 = 0.6$ and traces the expected welfare of each rule over (a, R_G^2) . The left panel orders the four leading rules. The oracle is, by construction, the upper envelope, and it increases with a because a larger a stretches the welfare landscape across treatments: when a is large, the best treatment is much better than the worst, so any rule that can find that treatment captures more value. The calibrated rule tracks the oracle most closely. The naive rule and the “ignore guardrail” rule lie strictly below it, and the gap between them is small when the guardrail is either very noisy or welfare-irrelevant.

The right panel of figure 4 plots the difference *calibrated* – *naive*. The geometry of this surface is the substantive content of the figure. The welfare advantage of calibration is essentially zero in two regions: (a) the front ridge where $a \approx 0$, since then the guardrail does not enter the rank no matter what one does with it; and (b) the right edge where $R_G^2 \approx 1$, since then there is no informational

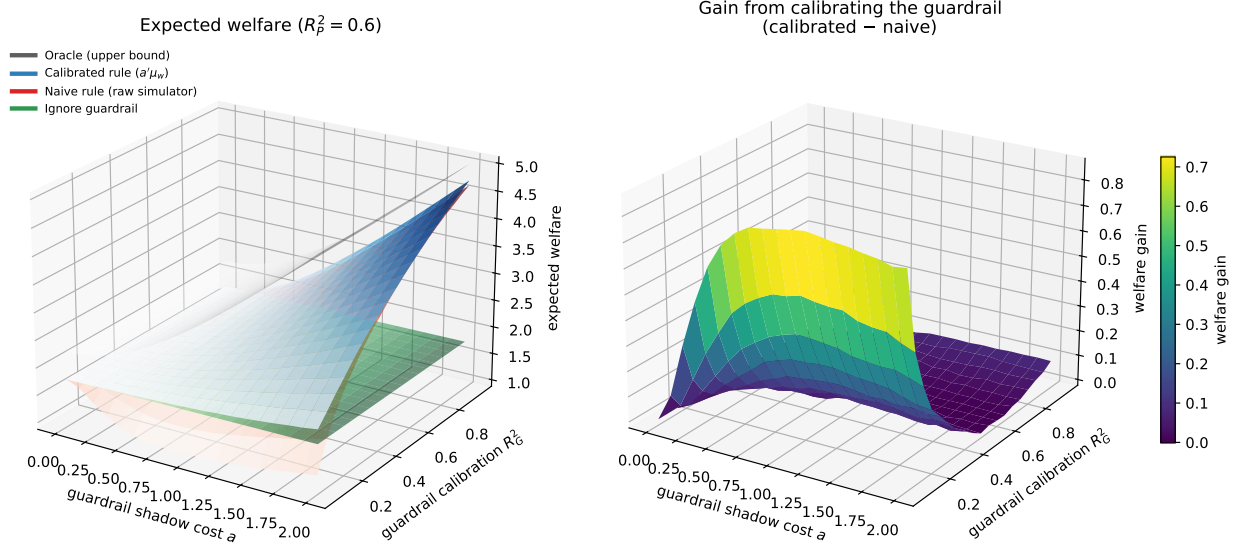


Figure 4: *Expected welfare and the value of calibration.* Left: expected realized welfare $\mathbb{E}[U(\theta_{w^*})]$ for four screening rules at primary calibration $R_P^2 = 0.6$, swept over the guardrail shadow cost a and the guardrail calibration R_G^2 . The oracle is the upper envelope; the calibrated rule from Corollary 1 tracks it most closely. Right: welfare gain of the calibrated rule over the naive rule. The gain peaks in the interior, where the guardrail simulator is informative enough to influence the rank but noisy enough that ignoring calibration misallocates slots.

gap between the raw simulator and the calibrated prediction. Both regions are predicted by the corollary on pivotality: the guardrail can affect the rank only when both $a_g \neq 0$ and the calibrated metric varies across treatments. The advantage of calibration is largest in the interior region of moderate a and moderate R_G^2 . Intuitively, this is the regime where the raw simulator is informative enough to influence ranking but noisy enough that using it without shrinkage misallocates testing slots: the naive rule chases simulator noise in the guardrail dimension and the calibrated rule does not.

G.3 Result 2: where should simulator-engineering effort go?

Figure 5 answers a budgeting question: given the current calibration of each simulator, is the next dollar of engineering effort better spent improving the primary simulator or the guardrail simulator? We fix the shadow cost at $a = 1$ and operationalize “improvement” as a +0.25 increase in the relevant R^2 , then compute, by common random numbers, the resulting welfare gain. The left panel plots the two marginal-gain surfaces. Both are decreasing in their respective R^2 , in the standard sense of diminishing returns to information. The right panel plots the difference *guardrail gain* – *primary gain*: red regions are where the primary simulator should be improved, blue where the guardrail simulator should be.

The structure of this surface is exactly what Corollary 1 predicts. The optimal direction of investment is governed by which axis has more slack relative to its welfare weight. When the primary simulator is already well calibrated (R_P^2 large, front of the plot) but the guardrail is not, the guardrail axis has both more headroom and a nonzero a_g , so the marginal welfare from improving the

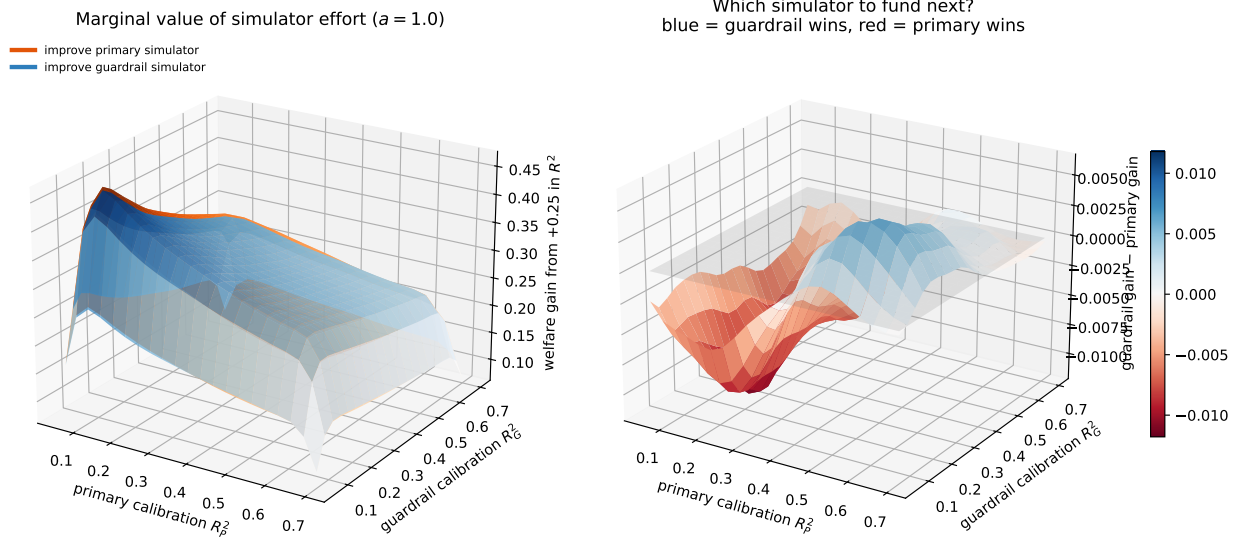


Figure 5: *Where to direct simulator effort.* Left: expected welfare gain from a $+0.25$ increase in R_P^2 (orange) versus R_G^2 (blue), at shadow cost $a = 1$. Right: difference *guardrail gain* $-$ *primary gain*. Blue regions identify (R_P^2, R_G^2) pairs where building a better guardrail simulator yields larger welfare than building a better primary simulator; red regions identify the opposite. The crossover line is approximately the iso-product $a^2 R_G^2 = R_P^2$ predicted by the local quadratic approximation.

guardrail simulator exceeds that of further refining the primary one. When the primary simulator is the worse-calibrated one, the order flips. The boundary between the two regions is, to a first approximation, the iso-product surface $a^2 R_G^2 = R_P^2$, which is the local welfare-variance argument from the LCB corollary applied at the simulator-effort margin.

This figure makes the policy point of Appendix E concrete. When the guardrail simulator is the comparative weakness of the simulator stack, it is also the one whose marginal calibration improvement is the most welfare-relevant. The screening theorem turns this into a screening rule: invest in the simulator whose product of welfare weight squared and remaining R^2 slack is largest.

G.4 Result 3: calibration can dominate policy relevance

The last result isolates the most counterintuitive implication of simulation weight = welfare sensitivity $\times$ calibrated informativeness. Figure 6 plots the welfare gain from including a guardrail metric in the calibrated rule, relative to the “ignore guardrail” rule, over the policy-relevance/calibration plane (a_g, R_G^2) . The 3D surface is a saddle-free monotone hill. The contour view on the right overlays the iso-curves of the effective weight $a_g R_G^2$.

Two illustrative points are marked. A guardrail metric that is highly policy-relevant ($a_g = 2.2$) but poorly calibrated ($R_G^2 = 0.1$) achieves an effective weight $a_g R_G^2 \approx 0.22$. A guardrail metric that is only modestly policy-relevant ($a_g = 0.6$) but well calibrated ($R_G^2 = 0.85$) achieves $a_g R_G^2 \approx 0.51$. The latter delivers a larger welfare gain than the former. In words: a manager who picks metrics for their face-value importance can be beaten by a manager who picks metrics for their measurability. The reason is exactly the corollary statement: it is $a_g \mu_{wg} = a_g R_g^2 \hat{\theta}_{wg}$ that enters the ranking score, and a poorly-calibrated metric contributes noise proportional to a_g rather than signal.

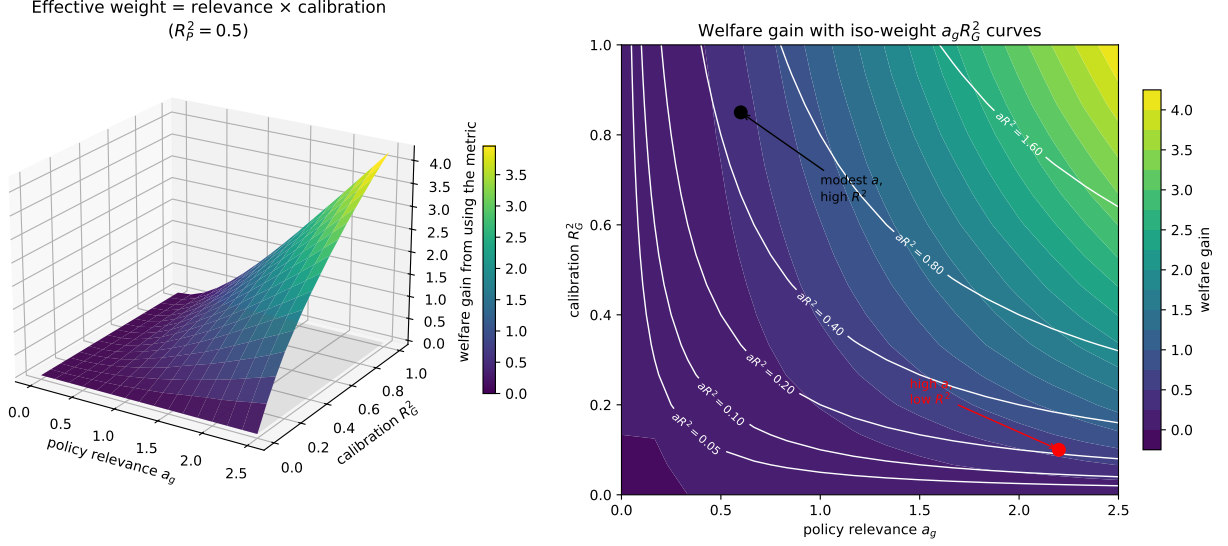


Figure 6: *Calibration vs. policy relevance*. Left: welfare gain from using the guardrail metric in the calibrated rule, relative to ignoring it, as a function of policy relevance a_g and calibration R_G^2 ; the gain is approximately monotone in their product. Right: top-down view with white iso-curves of the effective weight $a_g R_G^2$ and two annotated points. The high- a_g /low- R_G^2 point lies on a lower iso-curve than the modest- a_g /high- R_G^2 point, illustrating that a well-calibrated metric of moderate relevance can dominate a poorly-calibrated metric of high relevance.

G.5 Discussion of Simulation Results

- (1) **The screening theorem is empirically operative.** The calibrated rule $a' \mu_w$ tracks the oracle closely whenever the simulator carries any cross-treatment information, and dominates the naive raw-simulator rule by a margin that is largest exactly in the interior of the policy-relevance/calibration plane.
- (2) **Simulator-investment policy is governed by an iso-product condition.** The marginal welfare value of additional simulator quality is proportional to a_k^2 times the remaining R^2 slack on metric k . Resources should flow to whichever simulator has the largest such product, not to whichever simulator is most accurate or most policy-salient in isolation.
- (3) **Calibration can substitute for relevance.** Because the score depends on $a_k R_k^2$, a metric with a_k twice as large but R_k^2 four times smaller is strictly worse for screening. Practically, the firm's metric-selection problem and its simulator-engineering problem are coupled: a well-calibrated metric of moderate policy weight is often a better screening instrument than a poorly-calibrated metric of high policy weight.

Together, these results sharpen the practical content of the boxed identity in Appendix F: simulator effort earns its welfare only through the product of policy relevance and calibrated informativeness, and that product, not either factor alone, is the right object on which to allocate engineering and metric-design decisions.